

Tutorial Window Function con DuckDB (1ª parte)

```
pip install duckdb
```

```
import duckdb

# Import jupysql Jupyter extension to create SQL cells
%reload_ext sql
# Initiating a DuckDB database
conn = duckdb.connect("BDD/WindowFunction.duckdb")
%sql conn --alias duckdb
```

```
conn.sql("show tables")
```

name
varchar
order_items
orders
products

Table products

```
# create table products and load data into it
conn.sql("CREATE TABLE products AS\
        SELECT * FROM read_csv_auto('StarVision/products.csv',
        header=True, sep=',')");
```

```
# query the table
conn.table("products").show()
```

product_id	product_name	model_year
list_price		
int64	varchar	int64
double		
1	Trek 820 - 2016	2016
379.99		
2	Ritchey Timberwolf Frameset - 2016	2016

749.99			
	3	Surly Wednesday Frameset - 2016	2016
999.99			
	4	Trek Fuel EX 8 29 - 2016	2016
2899.99			
	5	Heller Shagamaw Frame - 2016	2016
1320.99			
	6	Surly Ice Cream Truck Frameset - 2016	2016
469.99			
	7	Trek Slash 8 27.5 - 2016	2016
3999.99			
	8	Trek Remedy 29 Carbon Frameset - 2016	2016
1799.99			
	9	Trek Conduit+ - 2016	2016
2999.99			
	10	Surly Straggler - 2016	2016
1549.0			
	.	.	.
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	.	.	.
.			
	312	Electra Townie Commute 8D Ladies' - 2018	2018
699.99			
	313	Electra Townie Original 1 Ladies' - 2018	2018
449.99			
	314	Electra Townie Original 21D EQ Ladies' - 2018	2018
679.99			
	315	Electra Townie Original 21D Ladies' - 2018	2018
559.99			
	316	Trek Checkpoint ALR 4 Women's - 2019	2019
1699.99			
	317	Trek Checkpoint ALR 5 - 2019	2019
1999.99			
	318	Trek Checkpoint ALR 5 Women's - 2019	2019
1999.99			
	319	Trek Checkpoint SL 5 Women's - 2019	2019
2799.99			
	320	Trek Checkpoint SL 6 - 2019	2019
3799.99			
	321	Trek Checkpoint ALR Frameset - 2019	2019
3199.99			
	321 rows (20 shown)		
columns			

```
| 321 rows (20 shown)
columns |
```

Table orders

```
# create table orders and load data into it
conn.sql("CREATE TABLE orders AS\
```

```
SELECT * FROM read_csv_auto('StarVision/orders.csv', header=True,
sep=',')");
# query the table
conn.table("orders").show()
```

order_id int64	order_date timestamp
1	2016-01-01 00:00:00
2	2016-01-01 00:00:00
3	2016-01-02 00:00:00
4	2016-01-03 00:00:00
5	2016-01-03 00:00:00
6	2016-01-04 00:00:00
7	2016-01-04 00:00:00
8	2016-01-04 00:00:00
9	2016-01-05 00:00:00
10	2016-01-05 00:00:00
.	.
.	.
.	.
1606	2018-07-10 00:00:00
1607	2018-07-11 00:00:00
1608	2018-07-12 00:00:00
1609	2018-08-23 00:00:00
1610	2018-08-25 00:00:00
1611	2018-09-06 00:00:00
1612	2018-10-21 00:00:00
1613	2018-11-18 00:00:00
1614	2018-11-28 00:00:00
1615	2018-12-28 00:00:00
1615 rows (20 shown)	

Table `order_items`

```
# create table order_items and load data into it
conn.sql("CREATE TABLE order_items AS\
SELECT * FROM read_csv_auto('StarVision/order_items.csv',
header=True, sep=',')");
# query the table
conn.table("order_items").show()
```

order_id int64	product_id int64	discount double
1	20	0.2
1	8	0.07
1	10	0.05

1	16	0.05
1	4	0.2
2	20	0.07
2	16	0.05
3	3	0.05
3	20	0.05
4	2	0.1
.	.	.
.	.	.
.	.	.
1612	293	0.07
1612	60	0.1
1613	153	0.07
1613	283	0.05
1614	124	0.07
1614	159	0.07
1614	213	0.2
1615	197	0.2
1615	214	0.07
1615	182	0.2
4722 rows (20 shown)		3 columns

```
conn.sql("show tables")
```

name
varchar
order_items
orders
products

```
# Explicitly close the connection
conn.close()
```